Math 321 — Introduction to Number Theory

Homework 3: Divisibility, Primes, and Congruences

Name: ____________________ Date: __________

Hello world

Instructions. Show all work clearly. Justify each step. Write your solutions in the spaces provided; use the back of the page or attach extra sheets if needed.

1 Divisibility

Definition – Divisibility. *Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say a* divides *b, written $a \mid b$, if there exists an integer k such that $b = ak$. We also say that a is a **divisor** (or **factor**) of b, and that b is a **multiple** of a.*

Theorem – Division Algorithm. *For any integers a and b with $b > 0$, there exist unique integers q (the* quotient*) and r (the* remainder*) such that*

$$a = bq + r, \qquad 0 \leq r < b.$$

Problem 1. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Hint. Write $b = ak_1$ and $c = ak_2$ for some integers k_1, k_2. What can you say about $b + c$?

Problem 2. Use the Division Algorithm to find the quotient and remainder when $a = 2\,027$ is divided by $b = 37$. Verify your answer.

Problem 3. Prove that the square of any integer is of the form $3k$ or $3k + 1$ for some integer k.

Hint. Consider the three cases $n \equiv 0, 1, 2 \pmod 3$ and square each.

2 Prime Numbers and the Fundamental Theorem of Arithmetic

Definition – Prime and Composite. *An integer $p > 1$ is called **prime** if its only positive divisors are 1 and p itself. An integer $n > 1$ that is not prime is called **composite**.*

Fundamental Theorem of Arithmetic. *Every integer $n > 1$ can be written uniquely (up to the order of the factors) as a product of primes:*

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad p_1 < p_2 < \cdots < p_k, \quad a_i \geq 1.$$

Problem 4. Find the prime factorization of each of the following:

(a) 3 780

(b) 10 ! (i.e. 10 factorial)

(c) $2^{12} - 1$

Problem 5. Prove that there are infinitely many prime numbers.

Hint (Euclid's argument). Suppose there are only finitely many primes $p_1, p_2, \ldots, p_n$. Consider the number $N = p_1 p_2 \cdots p_n + 1$. Is N divisible by any p_i?

Problem 6. Let p be a prime and let $a, b \in \mathbb{Z}$. Prove that if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Hint. This is sometimes called *Euclid's Lemma.* If $p \nmid a$, what is $\gcd(p, a)$? Use Bézout's identity.

3 Greatest Common Divisor and the Euclidean Algorithm

Definition – GCD. *The* ***greatest common divisor*** *of two integers a and b, not both zero, denoted* $\gcd(a, b)$*, is the largest positive integer d such that $d \mid a$ and $d \mid b$.*

Bézout's Identity. *For any integers a and b, not both zero, there exist integers x and y such that*

$$\gcd(a, b) = ax + by.$$

Problem 7. Use the Euclidean Algorithm to compute $\gcd(252, 198)$. Then find integers x and y such that $252x + 198y = \gcd(252, 198)$.

Hint. Apply repeated division: $252 = 198 \cdot q_1 + r_1$, then $198 = r_1 \cdot q_2 + r_2$, and so on until the remainder is 0. Back-substitute to find x and y.

Problem 8. Prove that if $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.

Hint. By Bézout, write $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$. Multiply these two equations together.

4 Congruences

Definition – Congruence. *Let m be a positive integer. We say a is **congruent** to b **modulo** m, written*

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$.

Fermat's Little Theorem. *If p is a prime and $\gcd(a, p) = 1$, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Problem 9. Compute each of the following. Show your reasoning.

(a) $3^{100} \pmod{7}$

(b) $2^{340} \pmod{11}$

(c) The last two digits of 7^{999}.

Hint. For (a) and (b), use Fermat's Little Theorem. For (c), work modulo 100 and find the pattern of powers of 7.

Problem 10. Solve the linear congruence $17x \equiv 3 \pmod{29}$.

Hint. Find the inverse of 17 modulo 29 using the Extended Euclidean Algorithm or by noting $17 \cdot (\,?\,) \equiv 1 \pmod{29}$.

Bonus Challenge

Problem 11 (Bonus). Prove **Wilson's Theorem**: if p is a prime, then

$$(p-1)! \equiv -1 \pmod{p}.$$

Hint. Pair each element $a \in \{1, 2, \ldots, p-1\}$ with its multiplicative inverse $a^{-1} \pmod{p}$. Which elements are their own inverses?

End of Homework 3